Project Reflection

The final project for the *Sensing the World* Module gave us an opportunity to actively use the Iterative process and allowed us to apply the skills we've learned. During our first session, we've brainstormed as to what we wanted to do. While brainstorming can sometimes be time-consuming, it wasn't the case for our group. We had some difficulty at first coming up with an idea since the prompt was so flexible. To make it easier, we decided to look at the project requirements and go there. We selected components that were most familiar with and from there try to create something that would fit the prompt. Based on this, we were able to land on a Automated Light System.

From here we decided to split the work among us so that we'd always be advancing towards the end of the project. For example, someone worked on the code ahead of time so it could be tested immediately once our circuit was built. Another worked on wiring our breadboard, and another the actual surrounding structure that would hold it in place out of cardboard. Collaboration and assigning certain roles were an important aspect that we thought was necessary in order to keep everyone contributing to the project. Although, we managed to finish the actual building on the first day, we had problems with our switch. The overall idea of our design was to have a servomotor flip a light switch once a light sensor detected it was daytime. However, one problem we had is that the servo motor didn't press our switch.

This was a major problem because our entire idea rested on the servo being able to flip the light switch. We solved this by switching what type the switch we used. At first, we were using a button, which we switched to an on/off switch. This didn't completely fix the problem, but it was very close to actually flipping it. While thinking about how to solve the problem, we realized that we'd never screwed in there servo motor blade. Once we did this there was enough torque to flip the switch. Furthermore, we reinforced the switch by glueing it alongside the servomotor rather than onto cardboard so it wouldn't shift over time. The coding part of the project was not as time-consuming as AI helped to develop the majority of the code and we just had to give it the necessary info. However, something we noticed is that if we weren't specific enough, it'd create code that didn't function as we wanted. All it took was more precise prompts and context to get the appropriate code. The rain sensor was by far the most time-consuming, because we had to remake it which took almost an entire day of work. Once we actually had it, we didn't really know how we wanted to implement it, because the majority of our components were inside. We decided to

incorporate it in a way that would satisfy one of the potential cases of our design. Specifically, what if it was raining outside in the morning. Naturally, there'd be clouds so it'd be darker outside even though it's daytime. Our humidity sensor could detect this rise in humidity and lower the threshold for which the light sensor would need to record daytime and still turn on the lights, even if it was darker outside. While we were able to use the humidity sensor, on the other hand, it was very unreliable. Whenever we turned the system on and off, it would read as if it was reading. Due to time constraints, we were ultimately unable to resolve this and is something that would need to be worked on in the future.

I think the experience as a whole gave me a better idea of what it's like to work on a team and how that can make work both easier/harder. Easier because work can be split up between team members which leaves you only to one part. This makes it less stressful overall as you're only responsible for your part and that's it. However, it can be difficult if everyone doesn't communicate which can make some members work harder than others. I think me and everyone else in my group tried to help where they could, which made the overall process easier. Alongside this it was nice to see how a simple idea could be prototyped to create a proof of concept. While it can be complicated, we found building from the ground up prevented us from getting confused as to whether a problem was hardware or software. We built one part and then tested if it worked and kept doing this until we finished our project. This saved us a lot of time while also breaking the code into chunks so we could test what part wasn't working. The overall process helped us become more familiar with the iterative process and can be applied throughout the next module. I think that moving forward this experience will help me in developing my own personal projects. The lessons I learned throughout not only the final project, but the module as a whole. Skills from time-management, communication, and professional etiquette will carry on throughout my career here at Lehigh and outside of it.